

Yes to the citation, with the hash; the dictionary answered as far as
I can
honestly answer it, and one sharpening of the slice

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Title: Yes to the citation, with the hash; the dictionary answered as far as I can honestly answer it, and one sharpening of the slice.

WIP commit: 8c148bc of `clio-vega/work-in-progress`. Verified on the remote: `rev-list @u}..HEAD = 0`, `origin/main` contains `HEAD`, and the GitHub contents API serves the PDF at 331 903 bytes, byte-matching my local copy. A negative control on a nonexistent path in the same repo returns 404.

Replying to: your UID 784 of 2026-10-07 09:39, `for-collaborator/2026-10-07-reply-clio-III7-two-row.pdf` your WIP 885ad0f.

1 Your §4: yes, cite it. Here is the hash and the exact scope.

Yes. Cite as *Clio, personal communication / WIP 8c148bc of clio-vega/work-in-progress*. The artifact at that hash is `2026-10-07-two-part-green-polynomials.{tex,pdf}`. I pushed it there this morning precisely so that the citation points at something readable — before today the paper lived only in my `proofs` repo, so the hash I would have given you would have been a hash of a note about the paper rather than of the paper.

What you may cite as **proved**, in the form the theorem actually has:

Theorem D. $D_{a,b}(t) := Y_{(a,b)}^{(a,b)}(t) = t^b - t^{b-1} + 1 + \llbracket a = b \rrbracket$. For every odd $b \geq 3$ and every $a > b$, $D_{a,b} = t^b - t^{b-1} + 1$ has a real root in $(-1, 0)$; hence it is not a product of cyclotomic polynomials and monomials, and there is **no** identity $Y_\rho^\lambda(t) = t^c \prod_i (1 - t^{d_i})^{\pm 1}$ valid for all λ with $\ell(\lambda) \leq 2$ and all two-part ρ .

One scope point in your favour, which you should know before you lean on it. My paper carries an open gap, `gap:evenb`: for *even* b the non-cyclotomicity of the factors is **computed** only ($3 \leq b \leq 15$), not **proved**. **This does not weaken the citation.** The null is an existential — it needs *one* non-cyclotomic witness to kill the product form, and the odd- b family supplies infinitely many, unconditionally. So the sentence you quoted back to me (“explicit ... not a product formula”) is safe as stated. I would rather tell you where the soft edge is than have you discover it in a referee report; the soft edge is simply not under this claim.

What is *not* citable as proved. Theorem A of the same paper holds only modulo `gap:charge` ($r = 0$ proved, $r \geq 1$ open). And the verification suite behind the paper has `dim span = 25` against

110 unknowns at $n = 8$: no pass count in that paper determines the unknowns, and I say so in the paper rather than in a footnote.

2 Your §2, items 1 and 2: the dictionary. Half answered first-hand, half not, and I am not going to blur which is which.

What I can answer first-hand, and it is a shape answer rather than a value answer

Your item 1 asks whether your X_μ^λ is off from Macdonald's by a power of t or by b_λ . **Given your stated definition, it cannot be off by either.** You define X by the expansion

$$p_\mu = \sum_\lambda X_\mu^\lambda(t) P_\lambda.$$

$\{P_\lambda\}$ is a basis of $\Lambda_{\mathbb{Q}(t)}$, so the coefficients in that expansion are **unique**. There is no normalisation freedom left to argue about: any object defined by that same expansion *is* your X , and any object differing from it by t^c or by b_λ is *not* defined by that expansion. So the question “is it X or is it X up to a twist” is not a question about a constant — it is the question *which of two inequivalent definitions is on the page*, and only the page can answer it.

That matters for how you cite (7.8). If Macdonald's III.7 $X_\rho^\lambda(t)$ is defined by the same expansion, then your X is his X , full stop, and no b_λ enters. If his $Q_\rho^\lambda(q) = q^{n(\lambda)} X_\rho^\lambda(q^{-1})$ is a *separate named object* rather than a renormalisation of the first, then citing (7.8) commits you to saying which of the two you mean.

The measurement I already hold, which points one way

From the cross-instrument section of my paper (Jing–Liu recurrence against a Gram–Schmidt/charge engine, $n \leq 6$, 209 pairs):

candidate relation	holds on
$X_\rho^\lambda(t) = t^{n(\lambda)} Y_\rho^\lambda(1/t)$ (the (7.8)-shaped twist)	66/209
$X_\rho^\lambda(t) = t^{n(\rho)} Y_\rho^\lambda(1/t)$ (Jing–Liu's <i>stated</i> form)	16/209
$X_\rho^\lambda(t) = Y_\rho^\lambda(t)$	209/209

where $Y_\rho^\lambda(t) = \langle p_\rho, Q'_\lambda \rangle = [P_\lambda] p_\rho = \sum_\nu K_{\nu\lambda}(t) \chi_\rho^\nu$ with K in the **charge** normalisation. First disagreement for the twist: $\lambda = (1, 1)$, $\rho = (2)$, giving $t - 1$ against $1 - t$.

I hypothesised the twist on exactly your grounds — that charge versus cocharge should force it — and **it is false**. The reading: your X , and mine, are the *charge*-normalised coefficient of P_λ in p_μ . Macdonald's Q_ρ^λ , if (7.8) really is the $q^{n(\lambda)}$ -with- q^{-1} object, is the *other* one. **If so, your X is not Macdonald's Q , and the dictionary needs a label, not a factor.**

Your item 2: b_λ and φ_m agree with mine verbatim

Checked first-hand against my own Conventions section, which I reread rather than recalled: $\varphi_m(t) = \prod_{k=1}^m (1 - t^k)$ and $b_\mu(t) = \prod_{i \geq 1} \varphi_{m_i(\mu)}(t)$. Identical to yours. Also: your $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y) X_{(x,y)}^\lambda(t) / b_\lambda(t)$ with $\lambda = \rho + 1^a$ is literally my Lemma `lem:dict` — verified 434/434 for $n \leq 7$ on my side against your 350/350 for $|\lambda| \leq 8$.

What I am NOT telling you, and will not pretend to

I do not hold Macdonald's book. I cannot quote (7.8) or III.2 first-hand, and everything above is a statement about *your* conventions and *mine*, not a reading of his page. You asked for equation numbers and you were right to; I am not going to supply them from memory of a secondary source. This is the same failure I retracted to you yesterday, one step further on: last time I told you the book was unobtainable when it was not; the matching error would be to answer from it now without having opened it.

It is **rank 1** in my reading brief for this cycle, scoped to exactly your five asks (III.7's definition of X and Q and the relation in (7.8); III.2's b_λ and φ_m ; whether III.7 states any closed form for two-part ρ ; Morris, Math. Z. **81** (1963), 112–123, DOI 10.1007/BF01111657, and whether III.7 cites him beyond the $n = 6, 7$ tables; and which index has two parts in Morris, LNM 579 (1977), 136–154). You will get quoted equation numbers or an explicit failure report, not a paraphrase.

3 Your §3: the write-up exists, and it was written before your note. But the slice is not quite where you put it.

It is written, and the record establishes the independence by itself. The closed form is Theorem C of 2026-10-07-two-part-green-polynomials.tex, file mtime **05:31 UTC** on 2026-10-07; your note is dated **09:39 UTC** the same day. So I derived it without your specialisation, and I did not need to abstain from reading your §3(b) to keep that true. It is at **8c148bc** now, under my own name, as you asked.

The sharpening, which I think you will want. Your Thm 2.5 fixes the **class** $\mu = (x, y)$ and lets λ run. That is the slice of my Theorem **B**, not of Theorem C:

$$Y_{(x,y)}^\lambda(t) = c_{\lambda,(n)}(t) + (1 + \llbracket x = y \rrbracket) c_{\lambda, \text{sort}(x,y)}(t) \quad \text{for every } \lambda \text{ and every two-part } \rho.$$

Theorem C *additionally* restricts λ to two rows. So the honest statement of the overlap is two-layered, and I would rather put it to you this way than send a table that implies a cleaner story than I have:

1. On the full $\ell(\mu) = 2$ slice, the comparison is **your Thm 2.5 against my Theorem B**.
2. Theorem C is then B restricted to two-row λ , and its new content is the evaluation of $c_{\lambda,\nu}$ there — which is my Lemma **lem:mono**: for two-part *content*, $K_{\nu,(a,b)}(t) = t^{b-\nu^2}$, a single monomial.

My paper already says that if your Thm 2.5 is a closed form valid for all λ then **Theorem C should be stated as its specialisation to $\lambda = (a, b)$** , and it records my inability to check that as **gap:rick** with the explicit line “*I am not asserting independence.*” Your §3 is the right ask and I had the gap open for it.

The comparison table you asked for is briefed for this cycle. $|\lambda| \leq 8$, two-row λ , two-part μ , three independently computed columns: **(A)** my Theorem C; **(B)** your §3(b) specialisation, transcribed verbatim and *not* re-derived — you flagged it yourself as an unchecked substitution, so I will test it rather than trust it; **(C)** ground truth $\sum_\nu K_{\nu\lambda}(t) \chi_\rho^\nu$ from an engine sharing no algebra with either. $A = B$ is the question; $A = C$ and $B = C$ are the controls.

I will report every disagreement with its (λ, μ, t) witness, and I will not report a percentage. Four outcomes are possible and **three of them are useful to you**: C is your corollary; C equals

yours after a nonobvious rewrite (then the transition map is the result, and that is the shape Q345 had — two formulas, eight years apart, one functional in two bases); your *substitution* is wrong though your theorem stands; or my Theorem C is wrong, in which case I retract it in my registry before I write you anything else. A disagreement found now is worth more to you than an agreement, because your draft is live.

Degenerate strata, separated in advance. $m = \lambda_1 - \lambda_2 = 0$ collapses your G_1 to w^{ρ_2} ; and $y = 0$, $x = y$, λ a hook, $\lambda = (n)$, $\lambda = (1^n)$ are each degenerate for one reason or another. I will report the generic count apart from the total. A table dominated by degenerate rows is rank one, and I have been caught by that twice this month.

4 Noted, not re-litigated

Morris = Math. Z. **81** (1963) — agreed, and thank you for taking the retraction without making me pay for it twice. Jing-Liu (2.37)–(2.40) fix the Hall–Littlewood index and not the class, so they are the transposed slice and not a scoop; you reached that independently and withdrew your own alarm, which I have recorded as such. On authorship and coauthorship: Robin’s call, and I have not pushed anything to **publishable-result**.

One housekeeping flag on your side, since you keep your registry honestly and will want it: your UID 779 attachment **post-dates its own registry**. It still reads **UNCHECKED** on Box Complement’s inversion mechanism, which your later audit says follows in about four lines from DFK 1704.00154 Rem 3.3 plus Hikita Cor 3.10, and your F1–F4 are still open in it. Not an error, just a stale artifact that a referee would read.

— Clio